BYRON DELANEY JR

QUANTITATIVE FINANCE · DATA SCIENCE · APPLIED MATHEMATICS

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TECHNICAL SKILLS

Languages
Python, SQL, C / C++, TypeScript, R

Quant Finance
Modern Portfolio Theory, Monte Carlo, credit risk, options & volatility surfaces, VaR / CVaR, Sharpe

ML & Statistics
scikit-learn, XGBoost, SHAP, NumPy, pandas, Gibbs sampling

Data Engineering
ETL design, Databricks, Polars, Parquet, SQLite, data validation

Tools & Web
Git, Jupyter, Streamlit, Plotly, React / Next.js

HIGHLIGHTS

0.788 AUC
Credit-default model over 50,000+ loans, with SHAP explanations

8.2M / sec
Option contracts priced by the C++ analytics engine

5,000
Monte Carlo portfolios mapped to the efficient frontier

6 markets
SPX · VIX · SPY · QQQ · ES · OEX covered by kTRM

3 systems
Credit-risk, options-analytics & portfolio tools shipped end to end

EDUCATION

M.S. Mathematics
California State University, East Bay — In progress

B.A. Applied Mathematics
University of California, Berkeley — 2024

Selected coursework
Real & Complex Analysis · Probability Theory · Numerical Analysis · Abstract & Linear Algebra

FOCUS AREAS

Derivatives & volatility modeling · credit risk modeling · portfolio optimization · high-performance data pipelines · open-source quant tooling

LANGUAGES

English — native
Spanish — native

EXPERIENCE

MaritAIme *Solutions Architect* — Jun 2026 – Present
Remote | early-stage quantitative software startup

- Build quantitative systems end to end — data pipelines, model backends, and interactive dashboards — owning features from design through deployment.
- Ship Python and SQL tooling that ingests, validates, and models business data, replacing manual spreadsheet reporting and cutting turnaround time.
- Stand up Streamlit / Plotly dashboards that turn model output into decisions non-technical stakeholders can act on directly.

Pixonomi *Data Science Intern* — Jul 2024 – Jan 2025

- Built and validated machine-learning models (scikit-learn, XGBoost) on datasets of 50,000+ records, evaluated with AUC and precision / recall.
- Engineered features and used SHAP to make model predictions explainable to non-technical stakeholders.
- Automated data cleaning and ETL, removing repetitive manual work from the analysis workflow.

Hyve Solutions *Repair Engineer* — Feb 2025 – May 2026

- Diagnosed and resolved hardware faults at scale in a high-volume data-center environment; tracked throughput and quality metrics.
- Trained incoming technicians one-on-one and in groups, breaking complex processes into clear, repeatable steps.
- Flagged recurring failure patterns from repair data to help cut avoidable rework.

Independent Mathematics Tutor *Self-employed* — 2020 – Present

- Teach algebra through multivariable calculus, linear algebra, statistics, and SAT / ACT math; bilingual instruction in English and Spanish.
- Build custom study plans around each student's pace, carrying students from failing grades to solid finishes across a full academic year.

SELECTED PROJECTS

kTRM — Options Analytics Engine
Python · C++ · Polars · Databricks · Dash / Plotly
Processes 8.2M option contracts per second across SPX, VIX, SPY, QQQ, ES and OEX. Volatility-surface calibration and arbitrage detection, with a strike-level dashboard for live inspection.

Credit Risk Scoring & Loan-Default Prediction
Python · scikit-learn · XGBoost · SHAP
Loan-default classifier reaching 0.788 AUC and 85% precision on 50,000+ records, with SHAP explanations behind every prediction.

Stock Portfolio Analysis Pipeline
Python · pandas · NumPy · SQLite
Runs 5,000-allocation Monte Carlo simulations to map the efficient frontier (Modern Portfolio Theory), with API ingestion into a normalized SQLite store.